

Лабораторная работа № 6

**Настройка пропускной способности глобальной сети с помощью
Token Bucket Filter**

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Содержание

1	Цель работы	4
2	Задание	5
3	Теоретическое введение	6
4	Выполнение лабораторной работы	7
5	Выводы	14
	Список литературы	15

Список иллюстраций

4.1	Задание топологии	7
4.2	ifconfig на хостах	8
4.3	Запуск iperf3 на хостах	9
4.4	Ограничение скорости на конечных хостах	10
4.5	Ограничение скорости на коммутаторах	10
4.6	Объединение NETEM и TBF	11
4.7	Скрипт для воспроизводимого эксперимента	12
4.8	Скрипт для отрисовки графика	12
4.9	График изменения скорости передачи	13

1 Цель работы

Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

2 Задание

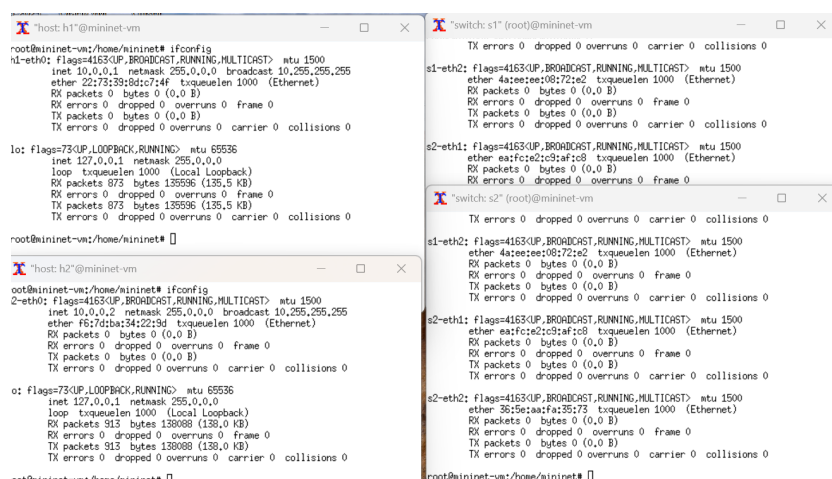
1. Задайте топологию, состоящую из двух хостов и двух коммутаторов с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по ограничению пропускной способности сети с помощью TBF в эмулируемой глобальной сети.
3. Самостоятельно реализуйте воспроизводимые эксперимент по применению TBF для ограничения пропускной способности. Постройте соответствующие графики.

3 Теоретическое введение

Mininet[**mininet?**] – это эмулятор компьютерной сети. Под компьютерной сетью подразумеваются простые компьютеры — хосты, коммутаторы, а так же OpenFlow-контроллеры. С помощью простейшего синтаксиса в примитивном интерпретаторе команд можно разворачивать сети из произвольного количества хостов, коммутаторов в различных топологиях и все это в рамках одной виртуальной машины(ВМ). На всех хостах можно изменять сетевую конфигурацию, пользоваться стандартными утилитами(`ifconfig`, `ping`) и даже получать доступ к терминалу. На коммутаторы можно добавлять различные правила и маршрутизировать трафик.

4 Выполнение лабораторной работы

Зададим простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8. На хостах h1 и h2 введем команду `ifconfig`, чтобы отобразить информацию, относящуюся к их сетевым интерфейсам и назначенным им IP-адресам. Проверим подключение между хостами сети (рис. 4.1).



```
root@mininet-vz:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 22:73:39:8d:c7:4f txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 873 bytes 135536 (135.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 873 bytes 135536 (135.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vz:/home/mininet#

root@mininet-vz:/home/mininet# ifconfig
2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether f8:76:b1:34:22:54 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 913 bytes 138088 (138.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 913 bytes 138088 (138.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vz:/home/mininet#

"switch: s1" (root@mininet-vz)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 4a:ee:ee:08:72:e2 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ea:fc:e2:c3:af:c8 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

"switch: s2" (root@mininet-vz)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 4a:ee:ee:08:72:e2 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

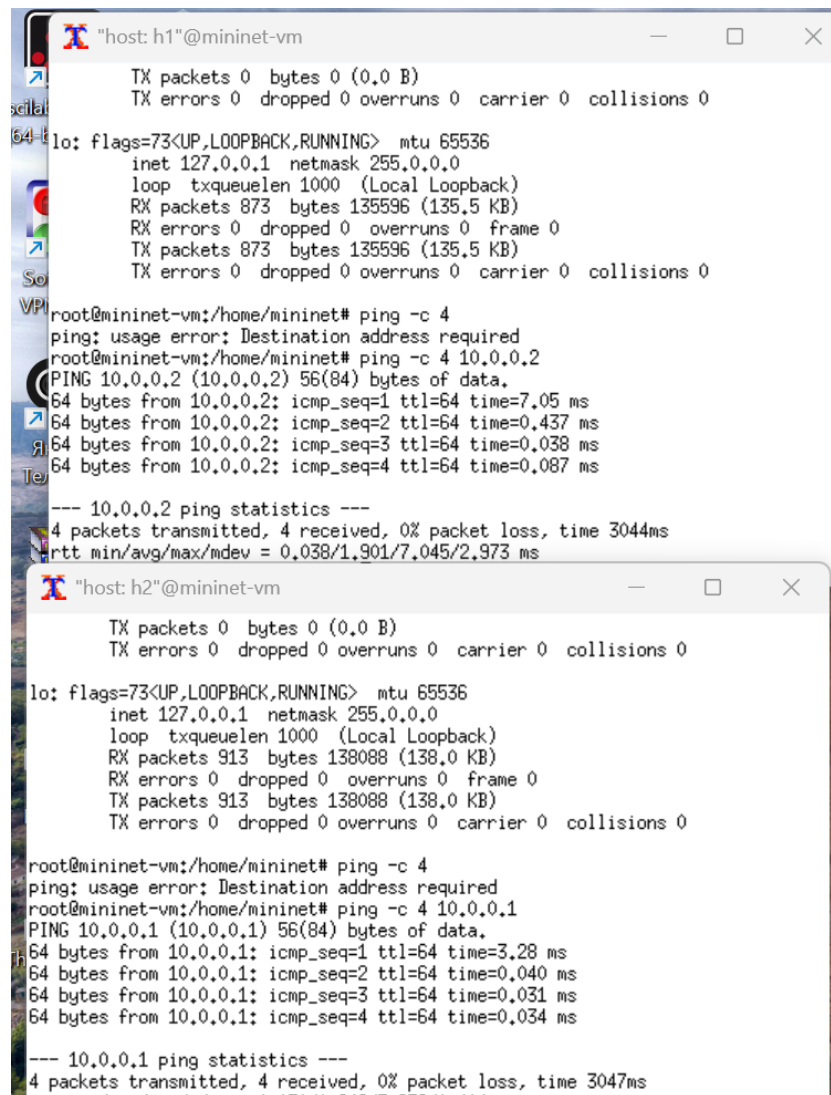
s2-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether ea:fc:e2:c3:af:c8 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 36:15e:aaf:35:73 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vz:/home/mininet#
```

Рис. 4.1: Задание топологии

На хостах h1, h2 и на коммутаторах s1, s2 введем команду `ifconfig`, чтобы отобразить информацию, относящуюся к их сетевым интерфейсам и назначенным им IP-адресам. В дальнейшем при работе с NETEM и командой `tc` будут использоваться интерфейсы h1-eth0, h2-eth0, s1-eth2 (рис. 4.2).



The image shows two terminal windows from a VMware environment. The top window is titled "host: h1"@mininet-vm and displays the output of the `ifconfig` command for the `lo` interface, showing it is configured with IP 127.0.0.1 and netmask 255.0.0.0. It also shows the output of a `ping -c 4 10.0.0.2` command, which successfully reaches the destination with 0% packet loss and a round-trip time of approximately 3044ms. The bottom window is titled "host: h2"@mininet-vm and shows the output of `ifconfig` for the `lo` interface with IP 127.0.0.1. It also shows the output of a `ping -c 4 10.0.0.1` command, which successfully reaches the destination with 0% packet loss and a round-trip time of approximately 3047ms.

```
"host: h1"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 873 bytes 135596 (135.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 873 bytes 135596 (135.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 4
ping: usage error: Destination address required
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=7.05 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.437 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.087 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3044ms
rtt min/avg/max/mdev = 0.038/1.901/7.045/2.973 ms

"host: h2"@mininet-vm
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 913 bytes 138088 (138.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 913 bytes 138088 (138.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 4
ping: usage error: Destination address required
root@mininet-vm:/home/mininet# ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=3.28 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.031 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.034 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3047ms
rtt min/avg/max/mdev = 0.034/0.040/3.280/0.004 ms
```

Рис. 4.2: ifconfig на хостах

Запустим iPerf3 на хостах и посмотрим результат отработки на данном этапе (рис. 4.3).

```

"host: h1"@mininet-vm
rtt min/avg/max/mdev = 0.038/1.901/7.045/2.973 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
iperf3: error - unable to connect to server: Connection refused
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 39612 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  5.59 GBytes 48.0 Gbits/sec  0   4.99 MBytes
[ 7] 1.00-2.00 sec  5.71 GBytes 49.1 Gbits/sec  0   4.99 MBytes
[ 7] 2.00-3.00 sec  5.53 GBytes 47.5 Gbits/sec 47   3.49 MBytes
[ 7] 3.00-4.00 sec  5.09 GBytes 43.7 Gbits/sec  1   3.49 MBytes
[ 7] 4.00-5.00 sec  4.89 GBytes 42.0 Gbits/sec  0   3.50 MBytes
[ 7] 5.00-6.00 sec  5.08 GBytes 43.6 Gbits/sec  1   3.50 MBytes
[ 7] 6.00-7.00 sec  5.62 GBytes 48.2 Gbits/sec  0   3.50 MBytes
[ 7] 7.00-8.00 sec  5.70 GBytes 48.9 Gbits/sec 47   2.90 MBytes
[ 7] 8.00-9.00 sec  5.50 GBytes 47.3 Gbits/sec  0   2.90 MBytes
[ 7] 9.00-10.00 sec 5.55 GBytes 47.6 Gbits/sec  0   2.90 MBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr
[ 7] 0.00-10.00 sec 54.3 GBytes 46.6 Gbits/sec  96
[ 7] 0.00-10.00 sec 54.3 GBytes 46.6 Gbits/sec
-----
sender
receiver

"host: h2"@mininet-vm
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 39608
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 39612
[ ID] Interval      Transfer    Bitrate      Retr
[ 7] 0.00-1.00 sec  5.59 GBytes 48.0 Gbits/sec
[ 7] 1.00-2.00 sec  5.71 GBytes 49.0 Gbits/sec
[ 7] 2.00-3.00 sec  5.53 GBytes 47.5 Gbits/sec
[ 7] 3.00-4.00 sec  5.09 GBytes 43.7 Gbits/sec
[ 7] 4.00-5.00 sec  4.89 GBytes 42.0 Gbits/sec
[ 7] 5.00-6.00 sec  5.08 GBytes 43.6 Gbits/sec
[ 7] 6.00-7.00 sec  5.62 GBytes 48.2 Gbits/sec
[ 7] 7.00-8.00 sec  5.70 GBytes 48.9 Gbits/sec
[ 7] 8.00-9.00 sec  5.50 GBytes 47.3 Gbits/sec
[ 7] 9.00-10.00 sec 5.55 GBytes 47.6 Gbits/sec
[ 7] 10.00-10.00 sec 3.62 MBytes 52.4 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.00 sec 54.3 GBytes 46.6 Gbits/sec
-----
receiver
Server listening on 5201
-----

```

Рис. 4.3: Запуск iperf3 на хостах

Изменим пропускную способность хоста h1, установив пропускную способность на 10 Гбит/с на интерфейсе h1-eth0 и параметры TBF-фильтра (рис. 4.4).

```

host: h1@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.087 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3044ms
rtt min/avg/max/ndev = 0.038/1.901/7.045/2.973 ms
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
iperf3: error - unable to connect to server: Connection refused
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 39612 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer     Bitrate      Retr  Cwnd
[ 7] 0.00-1.00    sec  5.59 GBytes  48.0 Gbits/sec  0  4.99 MBytes
[ 7] 1.00-2.00    sec  5.71 GBytes  49.1 Gbits/sec  0  4.99 MBytes
[ 7] 2.00-3.00    sec  5.53 GBytes  47.5 Gbits/sec  47  3.49 MBytes
[ 7] 3.00-4.00    sec  5.09 GBytes  43.7 Gbits/sec  1  3.49 MBytes
[ 7] 4.00-5.00    sec  4.88 GBytes  42.0 Gbits/sec  0  3.50 MBytes
[ 7] 5.00-6.00    sec  5.08 GBytes  43.6 Gbits/sec  1  3.50 MBytes
[ 7] 6.00-7.00    sec  5.82 GBytes  48.2 Gbits/sec  0  3.50 MBytes
[ 7] 7.00-8.00    sec  5.70 GBytes  48.9 Gbits/sec  47  2.90 MBytes
[ 7] 8.00-9.00    sec  5.50 GBytes  47.3 Gbits/sec  0  2.90 MBytes
[ 7] 9.00-10.00   sec  5.55 GBytes  47.6 Gbits/sec  0  2.90 MBytes
[ ID] Interval      Transfer     Bitrate      Retr  sender receiver
[ 7] 0.00-10.00   sec  54.3 GBytes  46.6 Gbits/sec  96
[ 7] 0.00-10.00   sec  54.3 GBytes  46.6 Gbits/sec

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc add dev hi-eth0 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 53860 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer     Bitrate      Retr  Cwnd
[ 7] 0.00-1.00    sec  1005 MBytes  8.43 Gbits/sec  0  3.37 MBytes
[ 7] 1.00-2.00    sec  1.10 GBytes  9.45 Gbits/sec  0  4.37 MBytes
[ 7] 2.00-3.00    sec  1.11 GBytes  9.58 Gbits/sec  135  4.59 MBytes
[ 7] 3.00-4.00    sec  936 MBytes  8.34 Gbits/sec  1260  3.30 MBytes
[ 7] 4.00-5.00    sec  633 MBytes  5.37 Gbits/sec  0  3.33 MBytes
[ 7] 5.00-6.00    sec  725 MBytes  6.08 Gbits/sec  45  2.44 MBytes
[ 7] 6.00-7.00    sec  1.11 GBytes  9.56 Gbits/sec  0  2.53 MBytes
[ 7] 7.00-8.00    sec  1.11 GBytes  9.57 Gbits/sec  0  2.75 MBytes
[ 7] 8.00-9.00    sec  1.11 GBytes  9.52 Gbits/sec  0  3.07 MBytes
[ 7] 9.00-10.00   sec  931 MBytes  7.81 Gbits/sec  0  3.08 MBytes
[ ID] Interval      Transfer     Bitrate      Retr  sender receiver
[ 7] 0.00-10.00   sec  9.75 GBytes  8.37 Gbits/sec  1440
[ 7] 0.00-10.00   sec  9.74 GBytes  8.36 Gbits/sec

host: h2@mininet-vm
Accepted connection from 10.0.0.1, port 39608
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 39612
[ ID] Interval      Transfer     Bitrate
[ 7] 0.00-1.00    sec  5.59 GBytes  48.0 Gbits/sec
[ 7] 1.00-2.00    sec  5.71 GBytes  49.0 Gbits/sec
[ 7] 2.00-3.00    sec  5.53 GBytes  47.5 Gbits/sec
[ 7] 3.00-4.00    sec  5.09 GBytes  43.7 Gbits/sec
[ 7] 4.00-5.00    sec  4.88 GBytes  42.0 Gbits/sec
[ 7] 5.00-6.00    sec  5.08 GBytes  43.6 Gbits/sec
[ 7] 6.00-7.00    sec  5.82 GBytes  48.2 Gbits/sec
[ 7] 7.00-8.00    sec  5.70 GBytes  48.9 Gbits/sec
[ 7] 8.00-9.00    sec  5.50 GBytes  47.3 Gbits/sec
[ 7] 9.00-10.00   sec  5.55 GBytes  47.6 Gbits/sec
[ ID] Interval      Transfer     Bitrate
[ 7] 0.00-10.00   sec  54.3 GBytes  46.6 Gbits/sec
receiver
Server listening on 5201

"CiperF3: interrupt - the server has terminated
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPV6 - trying IPV4
Server listening on 5201
Accepted connection from 10.0.0.1, port 53850
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 53860
[ ID] Interval      Transfer     Bitrate
[ 7] 0.00-1.00    sec  984 MBytes  8.34 Gbits/sec
[ 7] 1.00-2.00    sec  1.10 GBytes  9.45 Gbits/sec
[ 7] 2.00-3.00    sec  1.11 GBytes  9.58 Gbits/sec
[ 7] 3.00-4.00    sec  936 MBytes  8.33 Gbits/sec
[ 7] 4.00-5.00    sec  638 MBytes  5.36 Gbits/sec
[ 7] 5.00-6.00    sec  727 MBytes  6.10 Gbits/sec
[ 7] 6.00-7.00    sec  1.11 GBytes  9.56 Gbits/sec
[ 7] 7.00-8.00    sec  1.11 GBytes  9.56 Gbits/sec
[ 7] 8.00-9.00    sec  1.11 GBytes  9.53 Gbits/sec
[ 7] 9.00-10.00   sec  930 MBytes  7.80 Gbits/sec
[ 7] 10.00-10.00   sec  1.37 MBytes  4.15 Gbits/sec
[ ID] Interval      Transfer     Bitrate
[ 7] 0.00-10.00   sec  9.74 GBytes  8.36 Gbits/sec
receiver
Server listening on 5201

```

Рис. 4.4: Ограничение скорости на конечных хостах

Применим правило ограничения скорости tbf с параметрами rate = 10gbit, burst = 5,000,000, limit= 15,000,000 к интерфейсу s1-eth2 коммутатора s1, который соединяет его с коммутатором s2 (рис. 4.5).

```

switch: s1 (root@mininet-vm)
s1-eth2: Flags:4153<B, BROKDCST, RUNNING, MULTICAST> mtu 1500
ether 4a:ee:ee:08:72:e2 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth1: Flags:4153<B, BROKDCST, RUNNING, MULTICAST> mtu 1500
ether ea:fc:e2:c3:af:c8 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s2-eth2: Flags:4153<B, BROKDCST, RUNNING, MULTICAST> mtu 1500
ether 36:5e:a1:fa:35:73 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate
it burst 5000000 limit 15000000

host: h1@mininet-vm
Connecting to host 10.0.0.2, port 5201
[ ID] local 10.0.0.1 port 42296 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer     Bitrate      Retr  Cwnd
[ 7] 0.00-1.00    sec  936 MBytes  8.36 Gbits/sec  64  2.45 MBytes
[ 7] 1.00-2.01    sec  572 MBytes  4.75 Gbits/sec  0  2.49 MBytes
[ 7] 2.01-3.02    sec  429 MBytes  3.55 Gbits/sec  107  2.57 MBytes
[ 7] 3.02-4.00    sec  439 MBytes  3.77 Gbits/sec  225  1.36 MBytes
[ 7] 4.00-5.00    sec  531 MBytes  4.45 Gbits/sec  0  2.12 MBytes
[ 7] 5.00-6.00    sec  395 MBytes  3.32 Gbits/sec  158  2.16 MBytes
[ 7] 6.00-7.00    sec  434 MBytes  3.64 Gbits/sec  45  1.58 MBytes
[ 7] 7.00-8.00    sec  275 MBytes  2.30 Gbits/sec  90  1.68 MBytes
[ 7] 8.00-9.01    sec  362 MBytes  3.00 Gbits/sec  0  1.77 MBytes
[ 7] 9.01-10.00   sec  754 MBytes  6.42 Gbits/sec  135  1.80 MBytes
[ ID] Interval      Transfer     Bitrate      Retr  sender receiver
[ 7] 0.00-10.00   sec  5.07 GBytes  4.35 Gbits/sec  824
[ 7] 0.00-10.00   sec  5.06 GBytes  4.34 Gbits/sec

iperf Done.
root@mininet-vm:/home/mininet#

```

Рис. 4.5: Ограничение скорости на коммутаторах

Объединим NETEM и TBF, введя на интерфейсе s1-eth2 коммутатора s1 задержку, джиттер, повреждение пакетов и указав скорость. Добавим второе правило на

коммутаторе s1, которое задаёт ограничение скорости с помощью tbf с параметрами rate=2gbit, burst=1,000,000, limit=2,000,000: и проверим (рис. 4.6).

```

switch: s1 (root@mininet-vn)

s1-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether ea:fc:e2:c5:af:c8 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
ether 36:5e:a4:fa:35:73 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vn/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate
it burst 5000000 limit 15000000
root@mininet-vn/home/mininet# sudo tc qdisc del dev s1-eth2 root
root@mininet-vn/home/mininet# sudo tc qdisc add dev s1-eth2 root handle 1:
m delay 10ms
root@mininet-vn/home/mininet# sudo tc qdisc add dev s1-eth2 parent 1: hand
: tbf rate 2gbit burst 1000000 limit 2000000

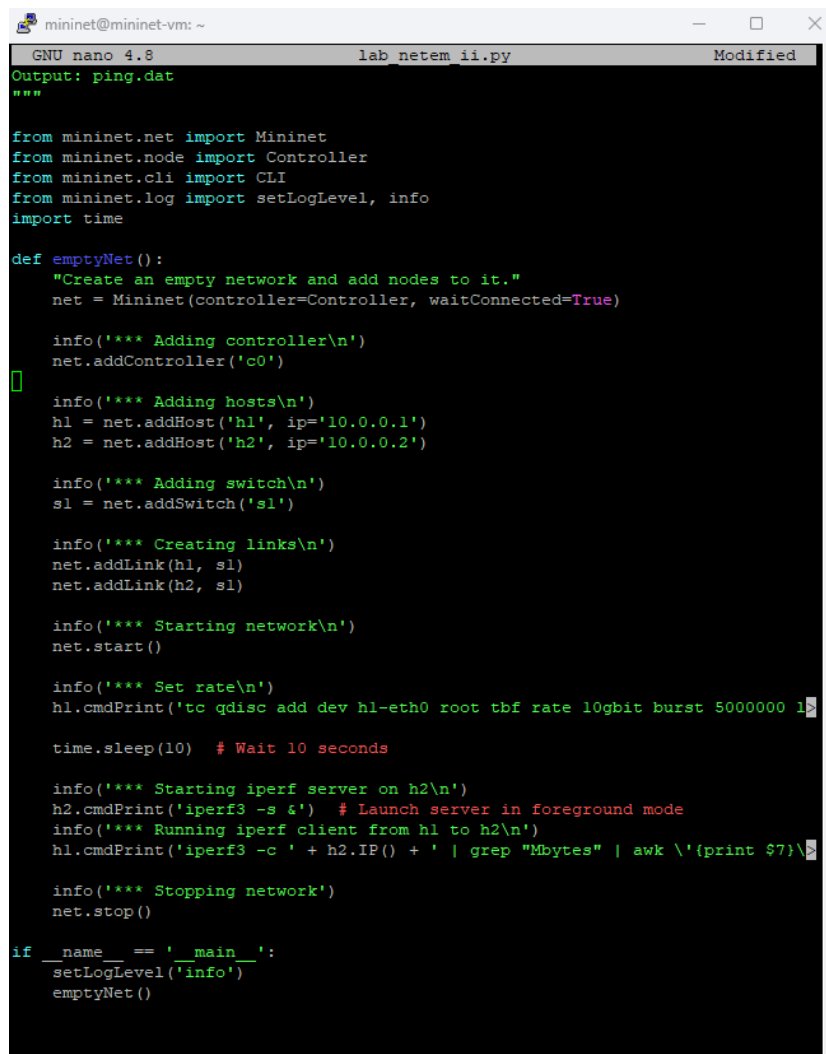
host: h1 (root@mininet-vn)

connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 57956 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer       Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec  60.5 MBytes  508 Mbits/sec  74    1.34 MBytes
[ 7] 1.00-2.04 sec  48.9 MBytes  392 Mbits/sec  37    1.40 MBytes
[ 7] 2.04-3.00 sec  26.2 MBytes  230 Mbits/sec  0     1.41 MBytes
[ 7] 3.00-4.00 sec  53.8 MBytes  452 Mbits/sec  0     1.44 MBytes
[ 7] 4.00-5.00 sec  37.5 MBytes  314 Mbits/sec  0     1.46 MBytes
[ 7] 5.00-6.00 sec  43.9 MBytes  367 Mbits/sec  7     1.48 MBytes
[ 7] 6.00-7.07 sec  27.5 MBytes  215 Mbits/sec  0     1.49 MBytes
[ 7] 7.07-8.06 sec  40.0 MBytes  338 Mbits/sec  0     1.51 MBytes
[ 7] 8.06-9.03 sec  23.8 MBytes  207 Mbits/sec  0     1.52 MBytes
[ 7] 9.03-10.00 sec  33.9 MBytes  310 Mbits/sec  0     1.57 MBytes
-----
[ ID] Interval      Transfer       Bitrate      Retr  sender receiver
[ 7] 0.00-10.00 sec  456 MBytes  382 Mbits/sec  118
[ 7] 0.00-10.01 sec  446 MBytes  374 Mbits/sec

perf Done.
root@mininet-vn/home/mininet#
eth2 root
  
```

Рис. 4.6: Объединение NETEM и TBF

В виртуальной среде mininet в своём рабочем каталоге с проектами создадим каталог simple-tbf и перейдем в него. Создадим скрипт для эксперимента lab_netem_iii.py (рис. 4.7).



```
mininet@mininet-vm: ~
GNU nano 4.8 lab_netem_i1.py Modified
Output: ping.dat
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.cli import CLI
from mininet.log import setLogLevel, info
import time

def emptyNet():
    "Create an empty network and add nodes to it."
    net = Mininet(controller=Controller, waitConnected=True)

    info('*** Adding controller\n')
    net.addController('c0')

    info('*** Adding hosts\n')
    h1 = net.addHost('h1', ip='10.0.0.1')
    h2 = net.addHost('h2', ip='10.0.0.2')

    info('*** Adding switch\n')
    s1 = net.addSwitch('s1')

    info('*** Creating links\n')
    net.addLink(h1, s1)
    net.addLink(h2, s1)

    info('*** Starting network\n')
    net.start()

    info('*** Set rate\n')
    h1.cmdPrint('tc qdisc add dev h1-eth0 root tbf rate 10gbit burst 5000000 1>

    time.sleep(10) # Wait 10 seconds

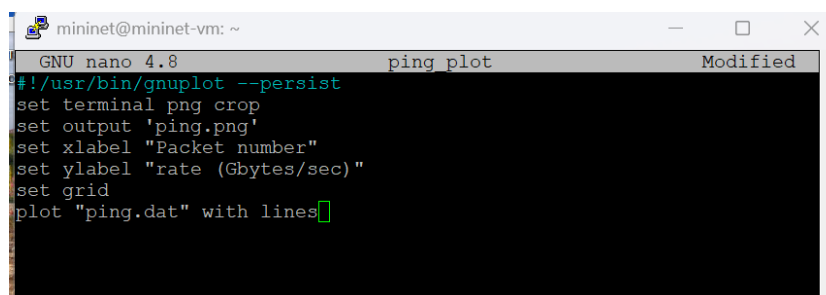
    info('*** Starting iperf server on h2\n')
    h2.cmdPrint('iperf3 -s &') # Launch server in foreground mode
    info('*** Running iperf client from h1 to h2\n')
    h1.cmdPrint('iperf3 -c ' + h2.IP() + ' | grep "Mbytes" | awk \'{print $7}\>

    info('*** Stopping network')
    net.stop()

if __name__ == '__main__':
    setLogLevel('info')
    emptyNet()
```

Рис. 4.7: Скрипт для воспроизводимого эксперимента

Создадим также скрипт для визуализации ping_plot результатов эксперимента (рис. 4.8).



```
mininet@mininet-vm: ~
GNU nano 4.8 ping_plot Modified
#!/usr/bin/gnuplot --persist
set terminal png crop
set output 'ping.png'
set xlabel "Packet number"
set ylabel "rate (Gbytes/sec)"
set grid
plot "ping.dat" with lines
```

Рис. 4.8: Скрипт для отрисовки графика

Получим следующий график (рис. 4.9).

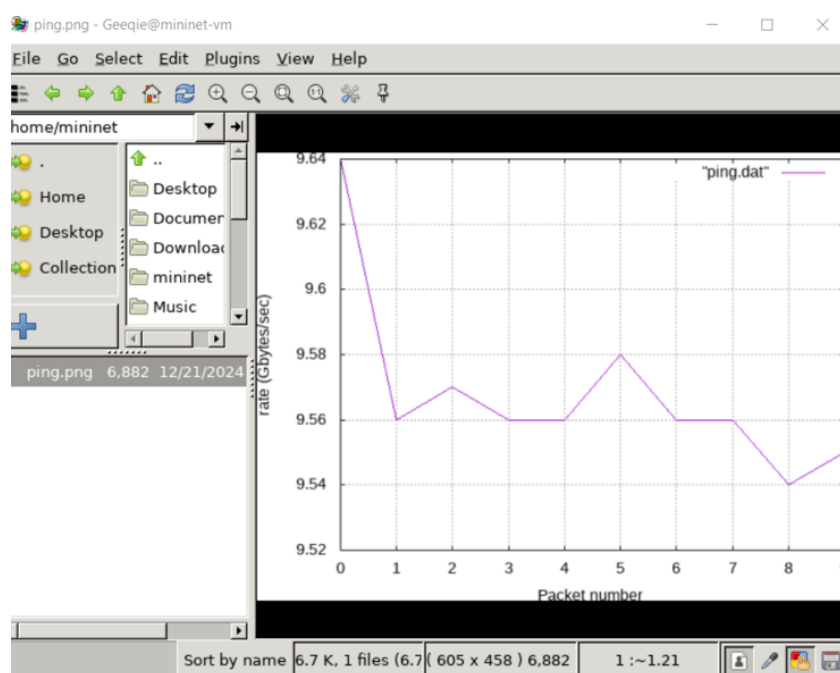


Рис. 4.9: График изменения скорости передачи

5 Выводы

В результате выполнения данной лабораторной работы я познакомился с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получил навыки моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

Список литературы